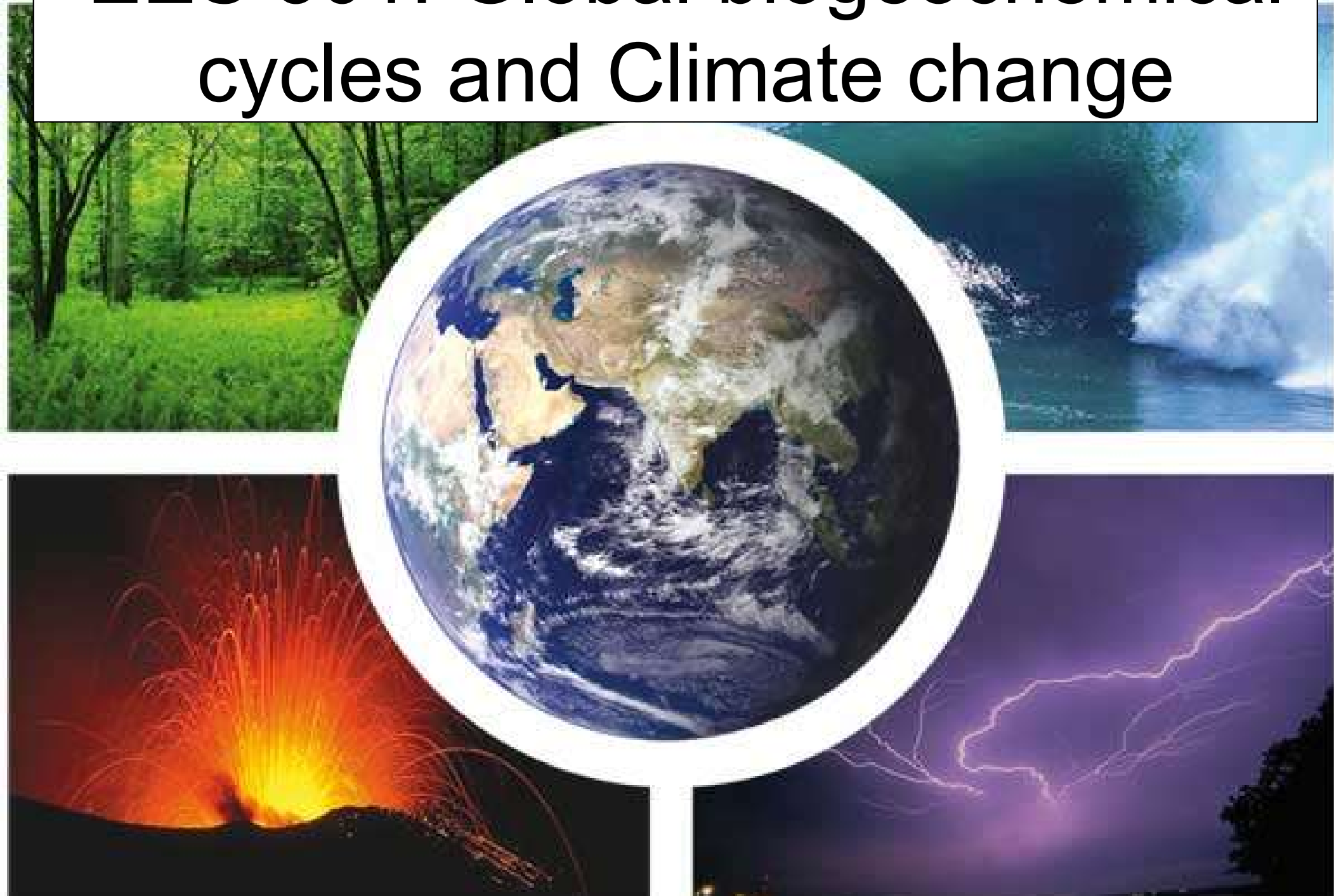


EES 301: Global biogeochemical cycles and Climate change



Thermodynamics and kinetics

- Very often, the box in a biogeochemical model is drawn for a particular chemical species and the flux from one box to the other happens in the form of some chemical reaction. This is why we will brush up on the chemical concepts required to describe such fluxes.
- Two fundamental concepts are very important in this context:
- **Thermodynamics describes whether a system is a equilibrium or can spontaneously change by undergoing chemical reaction**
- **Chemical kinetics describes the rate (speed) of the chemical change**

Limitations of applying thermodynamics and kinetics in natural systems

- Sometimes concepts of thermodynamics and kinetics can only be applied with some caution to natural systems due to several reasons:
- Some reaction rates can be so slow that natural systems stay in a non-equilibrium condition for a very long time period
- Natural catalysts, enzymes and surface chemistry are poorly understood at the moment
- Reactions in shrinking aerosol droplets can happen at mind boggling speed for reasons still not fully understood
- Nature is not a well mixed, continuously stirred beaker containing two chemicals only. When many substances compete e.g. for the OH radical things become too complex for a simple calculations
- There can be known unknowns and unknown unknown chemicals in your system

Laws of thermodynamics

- **Zeroth law:** If two systems are both in thermal equilibrium with a third system then they are in thermal equilibrium with each other. This law helps define the notion of temperature.

There exists a phenomenon called temperature, which is related to the kinetic energy inside the system

- **First law of thermodynamics:** Energy is neither created nor destroyed in a process but merely transformed from one form to another.

You can not create energy out of nowhere; the total energy contained in the universe is constant!

The best you can do is break even while shifting energy from one (useful) form into another (useful) form.

When the internal energy of your system decreases in a chemical reaction then the chemical change is accompanied by a release of energy often in the form of heat.

Laws of thermodynamics

- Second law of thermodynamics:** The entropy of a system and its surroundings increases for any spontaneous process ;

For any chemical process to be spontaneous, there must be an increase in entropy. You can think of entropy (S) as a measure of disorder.

You know this in real life: Your room becomes messy on its own (increase in entropy) but it takes energy to clean it up.

- Third law:** Absolute zero of temperature is approachable but unattainable .

Or

The entropy of a system approaches a constant value as the temperature approaches zero.

Or

Specifically, the entropy of a pure crystalline substance at absolute zero temperature is zero.

Important words to know while dealing with thermodynamics

- Enthalpy (H) is the sum of a thermodynamic system's internal energy (E) and the product of its pressure (p) and volume (V).

$$H=E+PV$$

It is a state function in thermodynamics which means it depends only on the state of the system and not on how it managed to arrive in that state.

- Temperature (T) reflects the average kinetic energy of the vibrating and colliding atoms making up a substance. It is a state function in thermodynamics.
- Pressure (symbol: p or P) is the force applied perpendicular to the surface of an object per unit area over which that force is distributed. It is a state function in thermodynamics.

Gibbs free energy

- Gibbs free energy is an important property of the system it is related to enthalpy and entropy by:

$$G = H - TS$$

- When the temperature of the reactants and products are equal ΔG is given by:

$$\Delta G = \Delta H - T\Delta S$$

- The second law of thermodynamics describes the equilibrium state of a system as that of maximum entropy and minimum free energy.

- For a system at constant temperature and pressure equilibrium conditions requires that the change in free energy is zero:

$$\Delta G = G_{\text{product}} - G_{\text{reactant}} = 0$$

$\Delta G = \Delta H - T\Delta S$ dissected

- The ΔH term is largely a function of the strength of the chemical bonds in the reactant and product. To the extent that there are more strong bonds in the product than in the reactant, ΔH will be negative. This contributes to the overall spontaneity of the reaction.
- The ΔS term is a measure of the relative degrees of disorder in reactant and product. To the extent that the product has greater disorder than the reactant ΔS will be positive. A positive ΔS contributes to the overall spontaneity of the reactions. ΔS can be evaluated from tables of entropy data.
- In practice G and H for a substance are defined relative to the G and H for the constituent elements of that substance. These values are known as free energy of formation and enthalpy of formation for standard conditions and symbolized by ΔG^0 and ΔH^0 . Values of these functions may be obtained from standard tables of thermodynamic data usually for the reference temperature of 298.2 K and a pressure of 1 bar. As a consequence of that definition elements in their standard state (e.g. N_2 gas or carbon as graphite) have free energies and free enthalpies of formation of zero.
- In the aqueous phase the change in G and H for the formation of an ion from its constituent element (e.g. $1/2 Cl_2(g) \rightarrow Cl^-(aq)$), the standard free energy or enthalpy of formation for ions, is defined relative to the change in G and H for the formation of H^+ from $H_2(g)$. As a consequence of this definition H^+ has zero free energies and free enthalpies of formation.

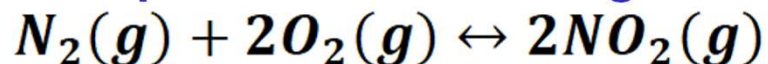
Gas phase equilibria

- For reactions in the gas phase the free energy per mol of gas as a function of pressure is expressed as:

$$G = G^0 + RT \ln(P/P^0)$$

with P^0 =standard pressure = 1 bar = 1/1.01325 atm, P = actual pressure of the gas, R = universal gas constant which has a value of 8.314 J/mol/K and T = temperature of the system in Kelvin

- For an ideal gas in equilibrium the change of the free energy for the reaction:



Is:

$$\Delta G = 2G_{NO_2} - G_{N_2} - 2G_{O_2}$$

We can substitute the topmost equation rearranged for G^0 . to get the equilibrium constant expression for gas phase reactions.

$$\Delta G^0 = -RT \ln \left(\frac{(P_{NO_2}/P^0)^2}{(P_{N_2}/P^0)(P_{O_2}/P^0)^2} \right) = -RT \ln K_P$$

This K_P constant is expressed in terms of partial pressures and unit less.

Gas phase equilibria

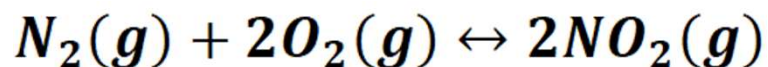
- In a mixture of gases, each constituent gas has a partial pressure which is the notional pressure of that constituent gas as if it alone occupied the entire volume of the original mixture at the same temperature

- Suppose for some reason you need K as K_C where each gas is being quantified in the units mol or molecules per unit volume then the conversion factor is:

$$K_C = K_P \left(\frac{P^0}{RT} \right)^{\Delta n}$$

Δn = change in the number of moles of gas during the reaction in this case

$\Delta n = -1$



Gas phase conversion of units

- Micrograms per Cubic Meter is related to the conserved unit parts per billion Volume ppbV often abbreviated as ppb in atmospheric chemistry as follows:

For an atmospheric concentration of NO₂ that is 50 ppb

$$\text{ppb NO}_2 = \frac{\text{molecules of NO}_2}{1 \text{ billion molecules of air}}$$

$$\text{ppb NO}_2 = \frac{\text{mol of NO}_2}{1 \text{ billion mol of air}}$$

If we want to convert that to microgram per meter cube, first we have to compute how many gram NO₂ there are in 50 ppb NO₂

Remember 12 g of ¹²C = 1 mol

$$\text{Therefore, } 46 \text{ g NO}_2 = \underline{1\text{-mol NO}_2}$$

Gas phase conversion of units

Next, using the ideal gas law, we can compute how many liters of air one mol of air occupies

$$\begin{aligned} V &= nRT/p = \frac{(1 \text{ mol}) (0.082 \text{ L atm mol}^{-1}\text{K}^{-1}) (T^{\circ}\text{K})}{(p \text{ atm})} \\ &= 24.6 \text{ L @ 300 K and 1 atm} \end{aligned}$$

This number is called the **molar volume**, it changes with both **pressure and temperature**. Calculating the correct molar volume is a step that carries its own marks in an evaluation.

$$\begin{aligned} (\text{g m}^{-3}) &= \text{Molecular weight (g)} \times \text{concentration (ppb)} \times 1000 (\text{L /m}^{-3}) / \text{Volume (L)} \\ \text{So } 46 \text{ g} \times 50 \times 10^{-9} \times 1000 (\text{L/m}^{-3}) / 24.6 \text{ L} &= 93.5 \times 10^{-6} \text{ g m}^{-3} = 93.5 \mu\text{g m}^{-3} \end{aligned}$$

If you don't worry about units just remember:

$$46 \times 50 / 24.6 = 93.5 \mu\text{g m}^{-3}$$

$$\underline{\text{Molecular weight (g)} \times \text{concentration (ppb)}} = \mu\text{g m}^{-3}$$

$$\text{Molar Volume (L)}$$

Gas phase conversion of units

Suppose we instead need how many molecules of NO₂ there are per cm³ of air

$$\begin{aligned} V &= nRT/p = \frac{(1 \text{ mol}) (0.082 \text{ L atm mol}^{-1}\text{K}^{-1}) (T^{\circ}\text{K})}{(p \text{ atm})} \\ &= 24.6 \text{ L @ 300 K and 1 atm} \end{aligned}$$

We need the molar volume, it changes with both pressure and temperature and Avogadro's number

$$(\# \text{ cm}^{-3}) = \text{Avogadro's number} \times \text{concentration (ppb)} \times 1 \text{ (L)} / 1000 \text{ (cm}^{-3}\text{)}/\text{Volume (L)}$$

$$\begin{aligned} \text{So } 6.022 \times 10^{23} \times 50 \times 10^{-9} \times 1 \text{ (L)}/1000 \text{ (cm}^3\text{)}/24.6\text{L} &= 6.022 \times 50 \times 10^{11} (\# \text{ cm}^{-3}) = \\ &\approx 300 \times 10^{11} (\# \text{ cm}^{-3}) \end{aligned}$$

Condensed phase equilibria

- For reactions in a pure liquid or solid the free energy per mol of gas as a function of pressure is expressed as:

$$G = G^0 + RT \ln(X)$$

With X as the mole fraction of the liquid.

- For reactions in dilute solution significantly less than 1M we use:

$$G = G^0 + RT \ln(C/C^0)$$

with the concentration units of the solute expressed in mol/L and C^0 as the concentration in the standard state defined exactly 1 molar (M)

The ratio C/C^0 is related to the activity for ideal solutions defined as

$$G = G^0 + RT \ln(\alpha)$$

And the change in free energy is expressed as: $\Delta G^0 = -RT \ln(K_{eq})$

Liquid units you should know

- **molarity = M = moles of solute / liters of solution**
- **Weight percent: When we say that ocean water contains 1.06% of Na⁺ we mean it contains 1.06g Na⁺ / 100 ml of solution**
- **If you need to convert such a number to molarity then the steps are: 1.06% = 10.6 g Na⁺ per 1 L**
- **10.6 g Na⁺ contains 10.6 g divided by 23 g/mol (molar mass of Na) mol of Na⁺ = 10.6 / 23 mol = 0.46 mol.**
- **The reversal of these steps may turn out useful if your guide gives you a salt like NaCl and tells you to prepare 0.1 molar solution**

Things you should not get confused about

- ppb and ppm are “nasty” units. People use them differently and rather loosely in different fields.
- People working on air **say ppm or ppb and always mean ppmV or ppbV** (parts per billion Volume) because that quantity much like partial pressure ratios does not change when you heat, cool, compress or decompress air. The oxygen percentage stays 21%.
- People working on water generally say ppm or ppb and mean mass of solute per unit volume of solution. Strictly speaking 1 ppm in water is 1 g of solute in 1 million g of water. When you use ppm the units in the numerator and denominator should be same because it is a unit less quantity! **However, people know that 1 g of pure water has a volume of 1 mL so they write 1 ppm = 1 g of solute in 1 million mL of water = 1 g of solute in 1000 L of water = 1 mg per L of water.** This conversion doesn't work if your water becomes dense, cold and saline in the ocean. It only works with dilute stuff in a chemistry lab.

Van't Hoff equation

- The van't Hoff equation describes the temperature dependence of the equilibrium constant.

$$\frac{d \ln K_{eq}}{dT} = \frac{\Delta H^0}{RT^2}$$

It can be seen that the degree of spontaneity at a given temperature depends on ΔG^0 but the change in spontaneity as defined by the temperature dependence of the equilibrium constant depends only on ΔH^0

The temperature dependence depends on the heat capacity of reactants and products.

Mixed phases

•The equilibrium constant can also be expressed when the reactants are in several phases. For example gas phase ammonia $\text{NH}_3(\text{g})$ will be in equilibrium with the dissolved ammonia $\text{NH}_3(\text{aq})$ in large cloud droplets

$$K_{eq} = \left(\frac{P_{\text{NH}_3}/P^0}{C_{\text{NH}_3}/C^0} \right)$$

Here P_{NH_3} is the ammonia partial pressure in air and C_{NH_3} is the concentration of undissociated ammonia in water

The equilibrium constant for this type of equilibria is often defined in terms of Henry's Law constant:

Note that K_H has units!

$$K_H = \left(\frac{P_{\text{NH}_3}}{C_{\text{NH}_3}} \right)$$

The solubility equilibrium between a solid and dissolved ions e.g. $\text{CaCO}_3 \leftrightarrow \text{Ca}^{2+}(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ is defined by:

$$K_{eq} = \left(\frac{C_{\text{Ca}^{2+}}}{C^0} \right) \left(\frac{C_{\text{CO}_3^{2-}}}{C^0} \right) \quad \text{or} \quad K_{SP} = [\text{Ca}^{2+}][\text{CO}_3^{2-}]$$

Aggregate Variables

- In a real world system the concentration of several species may be linked by chemical reactions that are a function of pH, oxidation potential or light intensity.
- Aggregate variables can be used to simplify the description of such complex systems.

• Let us look at this with an example Al(III) species in water include $AlOH^{2+}$, $Al(OH)_2^+$, $Al(OH)_3$ and Al^{3+} . Most of these species are amphoteric (which means they can react as acid or as base) and hence the exact speciation is extremely pH dependent. In this case the aggregate variable = total dissolved aluminum springs from the conservation of mass condition

$$[Al_T] = [Al^{3+}] + [AlOH^{2+}] + [Al(OH)_2^+] + [Al(OH)_3]$$

Carbonate and carbonic acid

- The carbonate equilibrium is of particular importance in the ocean and often treated with an aggregate variable. The total dissolved carbon C_T is represented by:

$$[C_T] = [CO_2(aq)] + [H_2CO_3] + [HCO_3^-] + [CO_3^{2-}]$$

- In the case of CO_2 one must consider both the hydrated form H_2CO_3 and the non hydrated form $CO_2(aq)$. The latter form is in equilibrium with the atmosphere:
 $CO_2(g) \leftrightarrow CO_2(aq)$

but $CO_2(aq)$ is in turn in equilibrium with H_2CO_3 : $H_2CO_3 \leftrightarrow CO_2(aq) + H_2O(l)$

The equilibrium constant for the carbonic acid equilibrium is $K_{eq} \approx 650$

The two species are often lumped together as carbonic acid or $H_2CO_3^*$

- The Henry's law and acid base equilibria are often written in terms of $H_2CO_3^*$
However, given the value of K_{eq} most "carbonic acid" is in fact $CO_2(aq)$

Alkalinity

- Alkalinity is helpful in describing the acid-neutralizing ability of an aqueous solution that may contain many ions:
- In any ionic equilibrium there is conservation of charge condition

$$\sum (+charges) = \sum (-charges)$$

- This can be broken down to

$$\begin{aligned} & \sum (+charges \text{ of conservative ions}) \\ & + \sum (+charges \text{ for nonconservative ions}) \\ & = \sum (-charges \text{ of conservative ions}) \\ & + \sum (-charges \text{ of nonconservative ions}) \end{aligned}$$

- Conservative ions are ones that do not undergo acid-base reactions at the pH value of interest. These include Na^+ , K^+ , Ca^{2+} , SO_4^{2-} , Cl^- etc.
- Nonconservative ions are ions that do undergo acid base reactions. These include H_3O^+ , OH^- , HCO_3^- and CO_3^{2-}

Alkalinity

- Alkalinity is defined as

$$\begin{aligned} \text{Alkalinity} &= \sum (+\text{charges of conservative ions}) \\ &- \sum (-\text{charges of conservative ions}) \\ &= \sum (-\text{charges of nonconservative ions}) \\ &- \sum (+\text{charges for nonconservative ions}) \end{aligned}$$

A 0.1 M solution of KOH has only one conservative ion K^+ , hence the alkalinity would be 0.1 M

A 0.1 M solution of HCl has Cl^- as the only conservative ion and its alkalinity is -0.1 M. Negative alkalinity is also known as acidity.

PRACTICE QUESTIONS

You can submit the answer to these questions by Monday 25th August if you want to get them corrected before the Midsem exam

Long answer style & numericals

1. Consider the reaction of $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \leftrightarrow 2\text{NO}_2(\text{g})$ for which $\Delta G^0_{298} = 103.68 \text{ kJ/mol}$ and $\Delta H^0_{298} = 67.70 \text{ kJ/mol}$
 - a) Evaluate K_p for this reaction at 298 K
 - b) For the atmosphere at sea level $P_{\text{O}_2} = 0.21 \text{ bar}$ and $P_{\text{N}_2} = 0.79 \text{ bar}$. Evaluate the pressure of NO_2 in equilibrium with these gases at 298K
 - c) Try to calculate the pressure of NO_2 for the tropopause $P_{\text{O}_2} = 0.021 \text{ bar}$ and $P_{\text{N}_2} = 0.079 \text{ bar}$ $T = 210 \text{ K}$
 - d) Evaluate K_p at 500 K assuming that ΔH^0 is independent of temperature
2. Evaluate the alkalinities of the following solutions
 - a) 0.1 M KHCO_3
 - b) 0.1 M K_2CO_3
 - c) 0.1 M $\text{Ca}(\text{OH})_2$
 - d) 0.1 M NH_4Cl

Long answer style & numericals

1. Convert the following mixing ratios of air pollutants from the units of ppbV to mass loadings in $\mu\text{g}/\text{m}^3$ and to molecules per cm^3 ($\# \text{ cm}^{-3}$)
 - a) 60 ppbV NO_2 at 298 K and 1 atm
 - b) 30 ppbV SO_2 at 278 K and 0.8 atm
 - c) 1 ppbV C_6H_6 at 315 K and 1.015 atm
2. Convert the following mass loadings in $\mu\text{g}/\text{m}^3$ of air pollutants to the units ppbV and molecules per cm^3
 - a) $2 \text{ mg}/\text{m}^3$ CH_4 at 305 K and 0.9 atm
 - b) $2 \text{ mg}/\text{m}^3$ CO at 280 K and 1 atm
 - c) $5 \mu\text{g}/\text{m}^3$ C_6H_6 at 295 K and 0.95 atm

Long answer style & numericals

1. The silver ion Ag^+ is an effective disinfectant for the water in swimming pools when it is kept at a concentration of 10 to 100 $\mu\text{g/L}$. A silver ion concentration higher than 300 $\mu\text{g/L}$ is considered to be unhealthy. One manufacturer of pool disinfectants provides a slightly soluble silver salt in pellet form that releases an equilibrium concentration of Ag^+ into the pool water. Which of the following silver salts would be the best candidate for this task:



2. List these three salts in order of decreasing solubility:

